

Analysis and Modeling on an Object Falling

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Abstract— In this study, the motion of an object is modeled with a second-degree polynomial based on erroneous height measurements related to the fall of an object from a height. The aim of this study is to reach accurate data with erroneous measurements and the model, and the object measured with erroneous measurements are simulated with a simulation developed using MATLAB. In addition, to better understand the modeling, the objects' position-time and velocity-time graphs are visualized instantly. The results provide the calculation of the initial velocity and gravitational acceleration and the formula of the equation of motion is presented. The simulation findings are consistent with theoretical calculations on this subject and this simulation provides the basis for correctly modeling erroneous measurements.

Keywords—MATLAB, simulation, least square method, interpolation, polynomial, initial speed, gravity acceleration, mathematical model

I. INTRODUCTION

The motion of objects plays an important role in understanding the dynamics of physical systems. In this context, erroneous height values of an object are given at certain times. The aim of this study is to examine the motion of the object with a second-degree polynomial model using this data. The least square method is used to model the second-degree polynomial model, and then interpolation is used to create the desired velocity-time and position-time graphs to better understand the model. Therefore, these concepts will also be explained in this study. Finally, it was requested to find the initial velocity of the object and the gravitational acceleration of the environment in which the object moves. At this point, a little more knowledge of mathematics and physics will be required.

II. METHODOLOGY

In this section, the least squares method and the concept of interpolation, which form the basis of the simulation, will be explained. If these are understood, the difficult part will be done because the rest of the code is largely graphics creation.

A. Least Square Method

The method of least squares is a technique used to construct a polynomial model that provides a good fit to a data set and a length or length that is used as a unit of relationships between a given dependent variable and one or more independent variables.

Generalizing k^{th} degree polynomial

$$y = a_0 + a_1x + a_kx^k \quad (1)$$

the residual is given by

$$R^2 \equiv \sum_{i=1}^n [y_i - (a_0 + a_1 x_i + \dots + a_k x_i^k)]^2. \quad (2)$$

The partial derivatives are

$$\begin{aligned} \frac{\partial(R^2)}{\partial a_0} &= -2 \sum_{i=1}^n [y_i - (a_0 + a_1 x_i + \dots + a_k x_i^k)] = 0 \\ \frac{\partial(R^2)}{\partial a_1} &= -2 \sum_{i=1}^n [y_i - (a_0 + a_1 x_i + \dots + a_k x_i^k)] x_i = 0 \\ \frac{\partial(R^2)}{\partial a_k} &= -2 \sum_{i=1}^n [y_i - (a_0 + a_1 x_i + \dots + a_k x_i^k)] x_i^k = 0. \end{aligned} \quad (3)$$

Partial derivatives lead to the following equations:

$$\begin{aligned} a_0 n + a_1 \sum_{i=1}^n x_i + \dots + a_k \sum_{i=1}^n x_i^k &= \sum_{i=1}^n y_i \\ a_0 \sum_{i=1}^n x_i + a_1 \sum_{i=1}^n x_i^2 + \dots + a_k \sum_{i=1}^n x_i^{k+1} &= \sum_{i=1}^n x_i y_i \\ a_0 \sum_{i=1}^n x_i^k + a_1 \sum_{i=1}^n x_i^{k+1} + \dots + a_k \sum_{i=1}^n x_i^{2k} &= \sum_{i=1}^n x_i^k y_i \end{aligned} \quad (4)$$

If these are written in matrix form:

$$\begin{bmatrix} n & \sum_{i=1}^n x_i & \dots & \sum_{i=1}^n x_i^k \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 & \dots & \sum_{i=1}^n x_i^{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n x_i^k & \sum_{i=1}^n x_i^{k+1} & \dots & \sum_{i=1}^n x_i^{2k} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \\ \vdots \\ \sum_{i=1}^n x_i^k y_i \end{bmatrix}. \quad (5)$$

This matrix form is called Vandermonde Matrix and can be obtained by writing the matrix for the fit of the least squares method:

$$\begin{bmatrix} 1 & x_1 & \cdots & x_1^k \\ 1 & x_2 & \cdots & x_2^k \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & \cdots & x_n^k \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \quad (6)$$

If we multiply the matrices at both ends of the equation by the transpose of the first matrix, the following will be obtained:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1^k & x_2^k & \cdots & x_n^k \end{bmatrix} \begin{bmatrix} 1 & x_1 & \cdots & x_1^k \\ 1 & x_2 & \cdots & x_2^k \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & \cdots & x_n^k \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1^k & x_2^k & \cdots & x_n^k \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad (7)$$

So,

$$\begin{bmatrix} n & \sum_{i=1}^n x_i & \cdots & \sum_{i=1}^n x_i^k \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 & \cdots & \sum_{i=1}^n x_i^{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n x_i^k & \sum_{i=1}^n x_i^{k+1} & \cdots & \sum_{i=1}^n x_i^{2k} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \\ \vdots \\ \sum_{i=1}^n x_i^k y_i \end{bmatrix}. \quad (8)$$

Under the fitting conditions of n points (x_i, y_i) given with polynomial coefficients, the following is obtained:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^k \\ 1 & x_2 & x_2^2 & \cdots & x_2^k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^k \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix}, \quad (9)$$

The following equation is given in matrix notation for polynomial fitting:

$$y = X a \quad (10)$$

The solution is continued by transposing the two sides:

$$X^T y = X^T X a \quad (11)$$

Then a is left alone in the equation and the solution is found:

$$a = (X^T X)^{-1} X^T y \quad (12)$$

We will find the coefficients of our model in the simulation with this method.

B. Interpolation

Since the data in the data set of the study is not very frequent, the values in between were found by interpolation. The process of obtaining missing data in a data group with the help of other data in the data group is called interpolation. There are many types of interpolation such as linear, curvilinear, cubic, etc. In this study, linear interpolation, which is the simplest method and suitable for the study, was used.

Let's explain linear interpolation: Linear interpolation is the method used to estimate the value of a point between two known values. In addition, linear interpolation is an effective method for curve fitting using linear polynomials. The linear interpolation formula is used for data estimation, data prediction, mathematical and scientific applications, and market research, etc. After the formula is given, a short example will be given to better understand:

$$(y) = y_1 + (x - x_1) (y_2 - y_1) / (x_2 - x_1) \quad (13)$$

Linear interpolation = (y) .

x_1 and y_1 are the first coordinates.

x_2 and y_2 are the second coordinates.

x is the point to perform the interpolation.

y is the interpolated value.

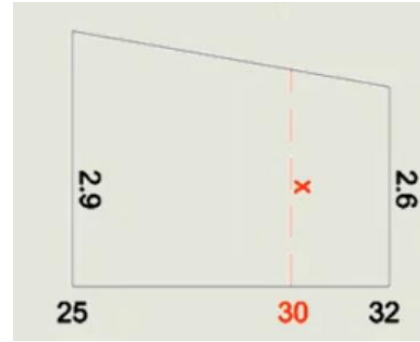


Fig. 1. Picture of example

If the x value is desired to be found by linear interpolation:
 $x = 2.6 + (2.9 - 2.6) (32 - 30) / (32 - 25)$ (14)

When the equation is solved, it is seen that x is 2.69.

The interpolation process in the simulation was performed with the `interp1` function, now the interpolation process in MATLAB will be explained with a simple MATLAB code to understand the interpolation process in the simulation.

Below are the temperature values at certain times.

time = 0, 1, 2, 3, 4, 5
temp = 0, 20, 60, 68, 77, 110

To find the temperature value at 3.6 seconds with the interpolation function:

It is sufficient to define the data set and write the x component you want to find together with the parameter names into the `interp1` function.

1	clc; clear all; close all
2	
3	x = 0:5;
4	y = [0 20 60 68 77 110];
5	
6	temp = interp1(x, y, 3.6)
7	

Command Window

temp =

73.4000

Fig. 2. Basic interpolation code and output

C. Explanation of some equations in the code

Here, it will be explained why the velocity of the modeled object, V_{pol} , is equal to $2a+b$, why the initial velocity is equal to the coefficient of the one-order term of the object, and why the gravitational acceleration of the environment is equal to $-2a$.

The equation of position for a constant acceleration object is:

$$x = x_0 + v_0 \cdot t + \frac{1}{2} a \cdot t^2 \quad (15)$$

x : Final position of the object (meter),

x_0 : Initial position (meter),

v_0 : Initial velocity (m/s),

a : Acceleration (m/s²),

t : Time (s).

This has the same format as the equation $at^2 + bt + c$.

Taking the derivative of the position equation gives the velocity equation, that is:

$$V_{pol} = d(at^2 + bt + c)/dt = 2a + b \quad (16)$$

To find the initial velocity, let's replace t with zero, leaving b alone. This explains why the initial velocity is equal to the coefficient of a degree term.

$$V(0) = 2a \cdot 0 + b = b \quad (17)$$

Finally, the gravitational acceleration is equal to $-2a$, which will be explained. If the two terms are examined carefully, the t^2 's should be equal, that is, $1/2g=a$. Of course, since the gravitational acceleration is directed towards the ground, it should be given a minus in front.

$$g = -2a \quad (18)$$

Thus, the equations to be used in the code were explained and confusion was prevented.

III. SIMULATION AND EXPLANATION

This section talks about performing the simulation. There can be many ways to perform the simulation, these ways depend on how expert you are in MATLAB. In this study, this simulation, which is done with introductory level knowledge, is examined. Of course, this study can be updated with the increase in knowledge in the following weeks. First, let's talk about the commands we used in the first three lines of the study. Initial settings:

The abbreviation of the "clc" command is "clear command window". This command clears the command window in MATLAB. It deletes previous outputs and commands in the command window, but memory and variables are not affected by this.

The abbreviation of the "close all" command is "close all figures". This command closes all open graphics in MATLAB. With this command, we can clear the graphics windows and make room for new graphics.

The abbreviation of the "clear all" command is "clear all variables". This command clears all variables in MATLAB, unlike the clc command, it also affects the memory and allows a new start by freeing the memory.

Then, the measured data are defined as column vectors. Then comes the section where the polynomial model is created.

Here, pol is the matrix required to create the polynomial using xi values. Since this polynomial is a parabola, it is a second-degree polynomial. (ax^2+bx+c)

Then, the coefficients of the polynomial are found with the mentioned least squares method. The coefficients are represented by a, b, c . Then comes the time change and polynomial value calculation part.

Time is divided into 300 equal intervals from 0 to 15. This time adjustment preserves the reality of the simulation. We calculate the height of the object we model by replacing the t 's in Y_{pol} with the time values we find. We set the negative values of the heights obtained by polynomial and interpolation to zero, because negative height is not meaningful. Then, the speed of the object being modeled is calculated using t and coefficients.

In this section, the basic derivative calculation and the interpolation process mentioned in the previous section are applied to the speed and height data of the measured object. If the derivative is mentioned: First, the V_{diff} series is defined. Then, the derivative is calculated by taking the differences between the y and x components classically. Then, the previously mentioned interpolation process is applied to the speed data and height data of the measured object. Then, just as we prevent the height of the model object from going to zero, we also prevent the height of the measured object from going negative here.

After these calculations, the section where they are turned into graphics is explained, there are 3 graphics, these are: a realistic simulation graphic of objects, speed-time and height-time graphics:

a new graphics window is started with figure. Afterwards, it is provided to open more than one graphics window with subplot. This graphic is the first graphic of the 1-row, 3-column graphic. We show the modeled object on the x-axis as a black circle at 1, 10 dimensions on the graph. We show the measured object on the x-axis as a red circle at 2, 10 dimensions on the graph. The boundaries of the graphic are: 0-3 in x, 0-350 in y. The name of the x component of the graphic is the location of the objects, and the name of the y component is the height of the objects. The graphic title was in a way that would allow the objects to be distinguished. Finally, there is the grid ten section that adds lines to the graphic so that the difference in movement between the objects can be understood more easily.

Then we move on to the height-time graph: the heights of the modeled object are represented by a black line with a thickness of 1. The given heights of the measured object are represented by a red line with a thickness of 1. The x component of the graph is called time, the y component is called height, and the title of the graph is height-time graph. Finally, there is the grid on section again.

Then we move on to the speed-time graph: the speeds of the modeled object are represented by a black line with a thickness of 1. The speeds of the measured object are represented by a red line with a thickness of 1. The x component of the graph is called time, the y component is called speed, and the title of the graph is speed-time graph. Finally, there is the grid on section again.

After the creation of the graphs section, the section of drawing the data in these graphic windows is explained:

The loop is started for all time steps with the for loop. Then, the heights of the homework object whose measurements are given for the first graph are set. If it is less than zero, set it equal to zero, otherwise update the height of the object measured according to time. Then, the height of the modeled object is updated in the same way. Then, the part for the second graph is explained: here, the x component of the graph is updated with time data. The y component is updated with height data. Finally, the part for the third graph is explained: here, the x component of the graph is updated with time data again. The y component is updated with height data. Then, we draw them instantly with the drawnow command and the pause command, which stops the simulation for 0.02 seconds, is added for the realism of the simulation.

Finally, the first speed found with mathematics, the gravitational acceleration and the printing part of the mathematical model will be explained.

The first speed is equal to the coefficient of the single-story element of the equation. The gravitational acceleration is equal to -2 times the two-story element, and the negative here is due to the gravitational acceleration being towards the ground. The mathematical model consists of the coefficients found. These are printed to the screen with fprintf

```

1      clc
2      close all
3      clear all
4
5      xi = [0; 1; 2; 3; 4; 5; 6; 7; 8; 9; 10; 11; 12; 13];
6      Yi = [202.36; 239.03; 280.71; 309.12; 323.15; 332.78; 328.45;
7           306.40; 287.36; 247.97; 202.89; 161.11; 93.68; 20.78];
8
9      pol = [xi.^2, xi, ones(length(xi), 1)];
10     coeffpol = (pol' * pol) \ (pol' * Yi);
11     a = coeffpol(1);
12     b = coeffpol(2);
13     c = coeffpol(3);
14
15     t = linspace(0, 15, 300);
16     Ypol = a * t.^2 + b * t + c;
17     Ypol = max(Ypol, 0);
18     Vpol = 2 * a * t + b;
19
20     Vdiff = zeros(1, length(Yi) - 1);
21     for i = 1:length(Vdiff)
22         Vdiff(i) = (Yi(i+1) - Yi(i)) / (xi(i+1) - xi(i));
23     end
24     Vhw = interp1(xi(1:end-1), Vdiff, t);
25     Yhw = interp1(xi, Yi, t);
26     Yhw = max(Yhw, 0);
27
28     figure;
29
30     subplot(1, 3, 1);
31     polobject = plot(1, Ypol(1), 'ko', 'MarkerSize', 10, MarkerFaceColor = 'k');
32     hold on;
33     hwobject = plot(2, Yhw(1), 'ro', 'MarkerSize', 10, MarkerFaceColor = 'r');
34     xlim([0, 3]);
35     ylim([0, 350]);
36     xlabel('location of objects');
37     ylabel('height of objects');

```

Fig. 2. Lines 1-37 of the simulation

```

38 title('pol object      /      hw object');
39 grid on;
40
41 subplot(1, 3, 2);
42 polheight = plot(t(1), Ypol(1), 'k', 'LineWidth', 1);
43 hold on;
44 hwheight = plot(t(1), Yhw(1), 'r', 'LineWidth', 1);
45 xlabel('time');
46 ylabel('height');
47 title('height-time graph');
48 grid on;
49
50 subplot(1, 3, 3);
51 polspeed = plot(t(1), Vpol(1), 'k', 'LineWidth', 1);
52 hold on;
53 hwspeed = plot(t(1), Vhw(1), 'r', 'LineWidth', 1);
54 xlabel('time');
55 ylabel('speed');
56 title('speed-time graph');
57 grid on;
58
59 for i = 1:length(t)
60
61     if Yhw(i) < 0
62         set(hwobject, 'YData', 0);
63     else
64         set(hwobject, 'YData', Yhw(i));
65     end
66
67     if Ypol(i) < 0
68         set(polobject, 'YData', 0);
69     else
70         set(polobject, 'YData', Ypol(i));
71     end
72
73     set(hwheight, 'XData', t(1:i), 'YData', Yhw(1:i));
74     set(polheight, 'XData', t(1:i), 'YData', Ypol(1:i));

```

Fig. 3. Lines 38-74 of the simulation

```

75
76     set(hwspeed, 'XData', t(1:i), 'YData', Vhw(1:i));
77     set(polspeed, 'XData', t(1:i), 'YData', Vpol(1:i));
78
79     drawnow;
80     pause(0.02);
81 end
82
83 v0 = b;
84 g = -2 * a;
85
86 fprintf('initial speed is %.4f m/s\n', v0);
87 fprintf('gravitational acceleration (g) is %.4f m/s^2\n', g);
88 fprintf('mathematical model of the motion is %.4f*t^2 + %.4f*t + %.4f\n', a, b, c);
89

```

Fig. 4. Lines 75-88 of the simulation

IV. RESULTS OF SIMULATION.

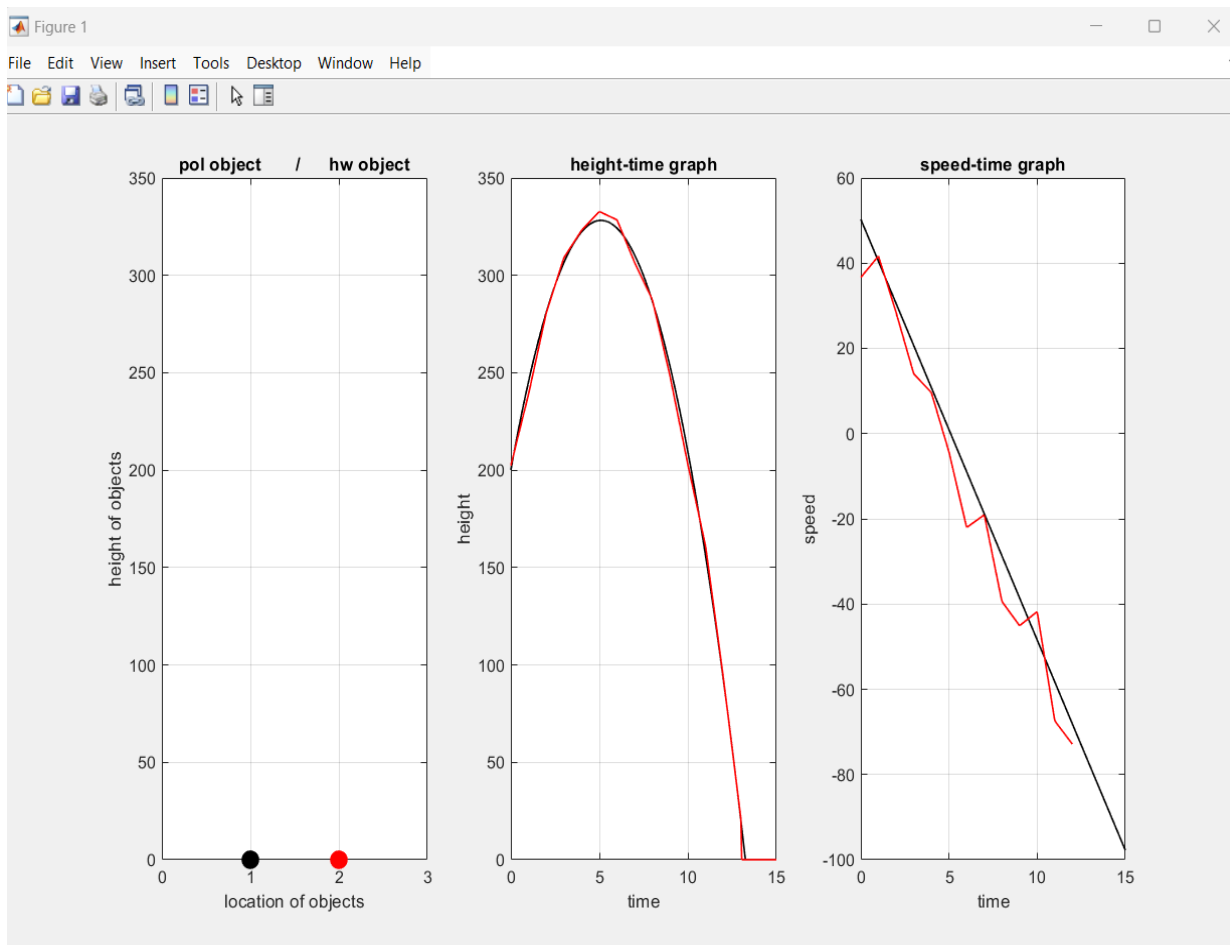


Fig.5 Output of code

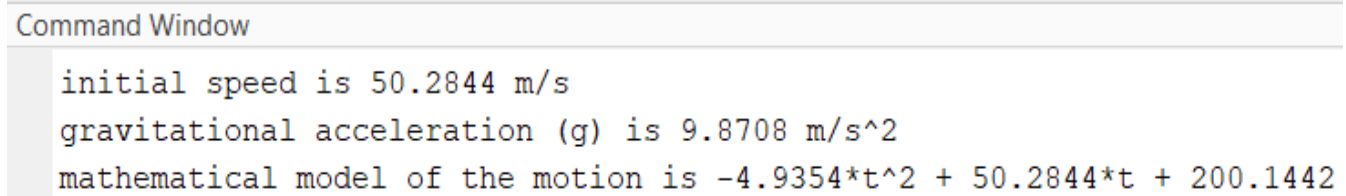


Fig. 6. Another output of code

As explained in the second section, firstly, the movements of the measured object and the modeled object were intended to be simulated side by side in a realistic way. When the results were examined, the first graph was a realistic simulation of these two objects. As explained in the second section, firstly, the height-time graphs of the measured object and the modeled object were requested and since some of the data were given incorrectly and with longer time intervals, it was expected that there would be differences between the graphs. When the simulation outputs were examined, these differences could be easily seen in the simulation outputs. Finally, as explained in the second section, the speed-time graphs of the measured object and the modeled object were requested, and the same difference was expected in the speed-time graph. When the results were examined, it could be seen that the expected differences occurred in this graph as well. Finally, the

simulation was asked to calculate the initial velocity, gravitational acceleration, and the mathematical model of the movement and print them on the screen. When the results were examined, it was seen that the simulation fulfilled these requests and printed them on the screen.

V. CONCLUSIONS OF STUDY

As a result, this study, which aims to model a free fall motion given erroneous measurements and to make the erroneous and modeled motion of this motion a realistic simulation and to graphically transform the data of the motions such as speed and height, has achieved its goal and is compatible with the predictions and equations. This simulation has become a basic model to see the effect of gravity on objects within the framework of Newtonian Mechanics and to understand the effect of modeling on the complex problem. Movements in larger dimensions and

different observation errors can make the simulation more complex. However, with the comprehensive mathematics and simulation features provided by MATLAB, it has become possible to solve and simulate such complex physics and mathematics problems. With the increase in the level of knowledge of MATLAB, it has been understood that it is a program that can be used in much different and much more complex problems. For example, it has been understood that modeling the horizontal projectile motion that we have seen in physics classes before can simulate tasks such as the rockets we produce performing more successful space missions. In this way, it has been concluded that such simulations that can be done in MATLAB can make significant contributions to technology and science over time.

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